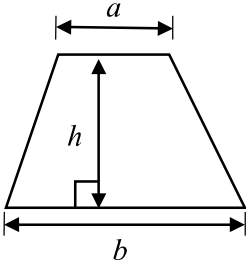
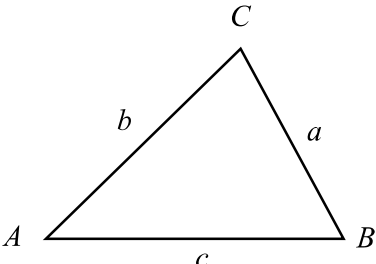
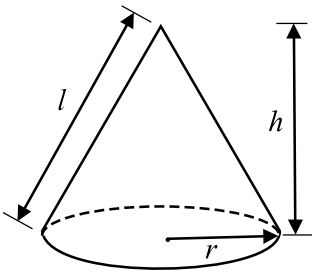
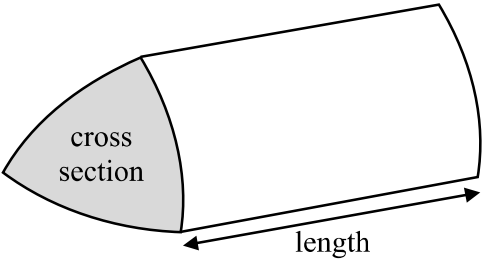
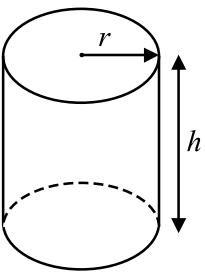


Appendix 5: Higher Tier formulae sheet

Arithmetic series Sum to n terms, $S_n = \frac{n}{2} [2a + (n-1)d]$ The quadratic equation The solutions of $ax^2 + bx + c = 0$ where $a \neq 0$ are given by: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	Area of trapezium $= \frac{1}{2}(a+b)h$ 
Trigonometry 	In any triangle ABC Sine Rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ Cosine Rule $a^2 = b^2 + c^2 - 2bc \cos A$ Area of triangle $= \frac{1}{2}ab \sin C$
Volume of cone $= \frac{1}{3}\pi r^2 h$ Curved surface area of cone $= \pi r l$ 	Volume of prism $= \text{area of cross section} \times \text{length}$ 
Volume of cylinder $= \pi r^2 h$ Curved surface area of cylinder $= 2\pi r h$ 	Volume of sphere $= \frac{4}{3}\pi r^3$ Surface area of sphere $= 4\pi r^2$ 